ALMA Proposal Preparation Workshop

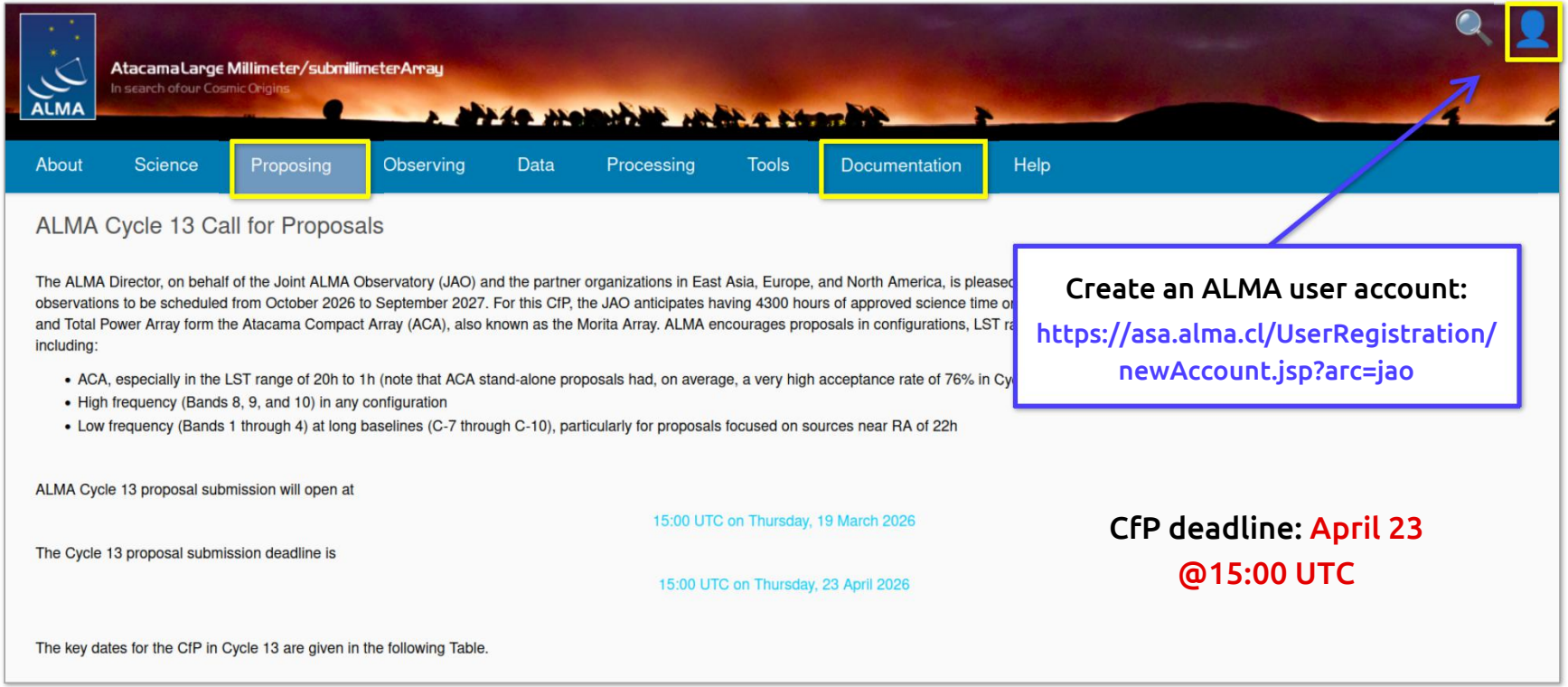
March 20, 2026 @IFA-UV, Valparaíso

2.6. ALMA Observing Tool (OT)

Carla Cornil-Baïotto

carla.cornil@postgrado.uv.cl

Cycle 13 Call for Proposal Portal



Create an ALMA user account:
https://asa.alma.cl/UserRegistration/newAccount.jsp?arc=jao

CfP deadline: April 23 @15:00 UTC

Cycle 13 CfP: https://almascience.nrao.edu/proposing/call-for-proposals

[Reminder] Proposal Preparation Overview

CfP deadline: April 23

- **Create an ALMA account**
  https://asa.alma.cl/UserRegistration/newAccount.jsp?arc=eu
  - Permanent username
  - Select expertise keywords
- **Observing Tool (OT)**
  https://cycle-13.sps.alma.cl/ngot/
  - Design the Science Goals in OT, test project feasibility
- **Project preparation**
  - Prepare the science case (SJ; must be strictly anonymous)
  - Justify technical setup (TJ)

Timeline

Key considerations

→ Sensitivity required
→ Angular resolution
→ Spectral resolution
→ Target size scales

Range of angular scales

$$\lambda/B_{max} < \theta < \lambda/B_{min}$$

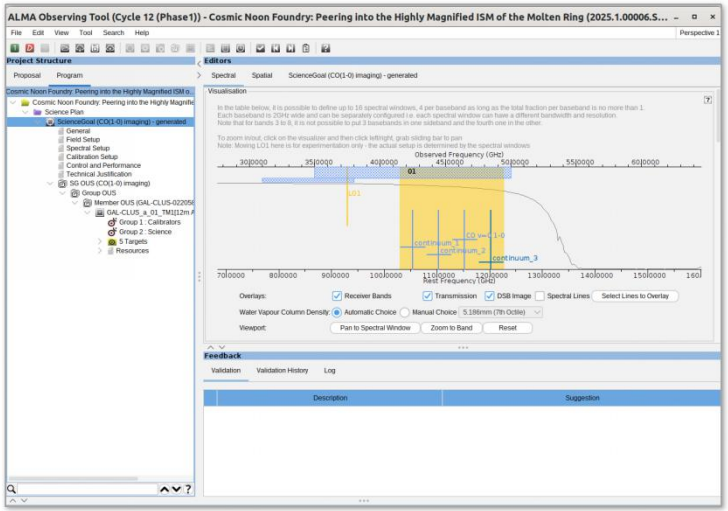
$$AR < \theta < MRS$$

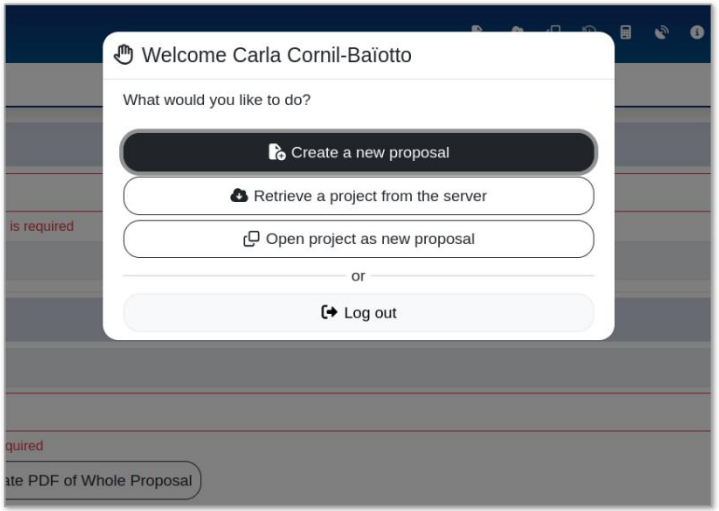
AR: Angular Resolution
MRS: Maximum Recoverable Scale

[Reminder] Cycle 13 Capabilities

Observing mode	12-m Array	7-m Array	TP Array
Spectral line imaging	Bands 1–10	Bands 1 and 3–10	Bands 1 and 3–10
Continuum imaging	Bands 1–10	Bands 1 and 3–10	Bands 1 and 3–10
Spectral scan mode	Bands 1–10	Bands 1 and 3–10	Bands 3–8
New Band 2 (67–116 GHz)	Yes	No	No
Mosaic observations	Bands 1–10	Bands 1 and 3–10	N/A
Polarization	Full polarization (linear + circular), Bands 1–7	Linear only, Bands 1 and 3–7	Not supported
Band-to-band phase calibration (B2B)	Available (Bands 7–10, conditional on calibrator)	Available (Bands 7–10, conditional on calibrator)	Not available

ALMA Observing Tool (OT)





Transition from local to web-based version

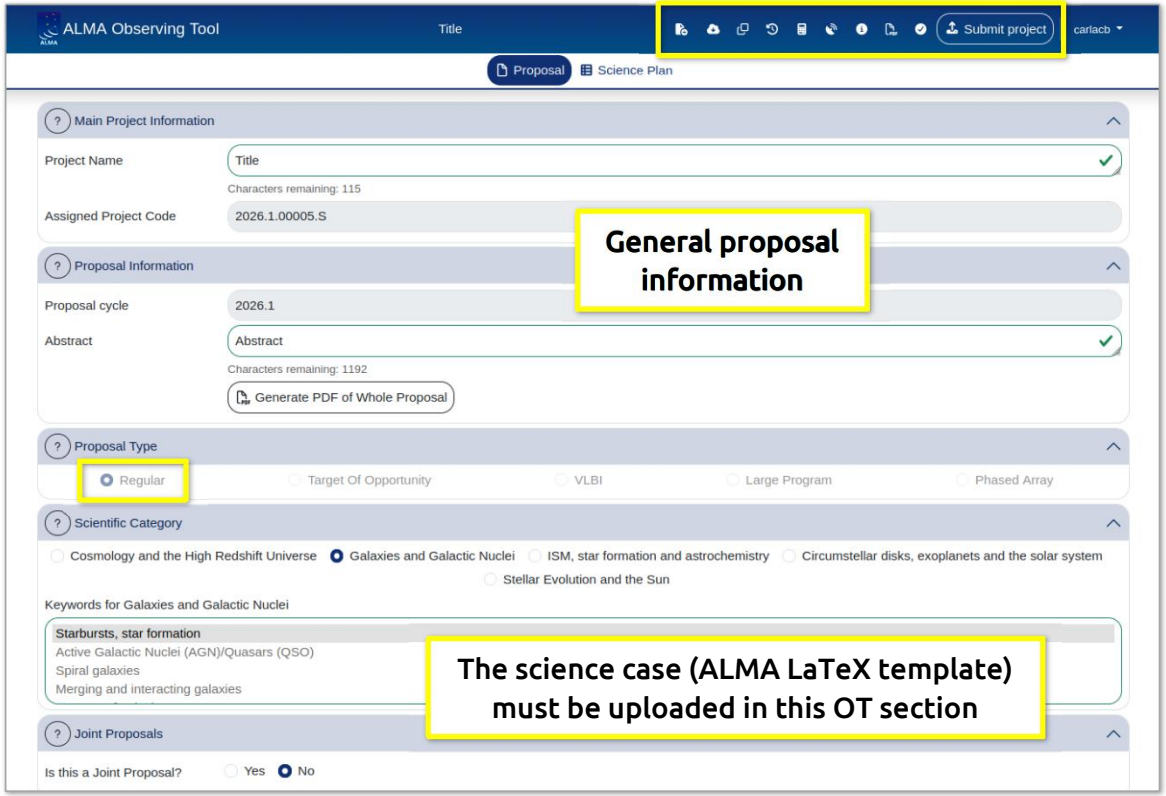
ngOT: https://cycle-13.sps.alma.cl/ngot/

➤ Short ALMA OT demo

Two sections:

1) Proposal
2) Science Plan

New Web-Based OT



- **Proposal section**

  https://almascience.nrao.edu/proposing/proposers-guide#autotoc-item-autotoc-42

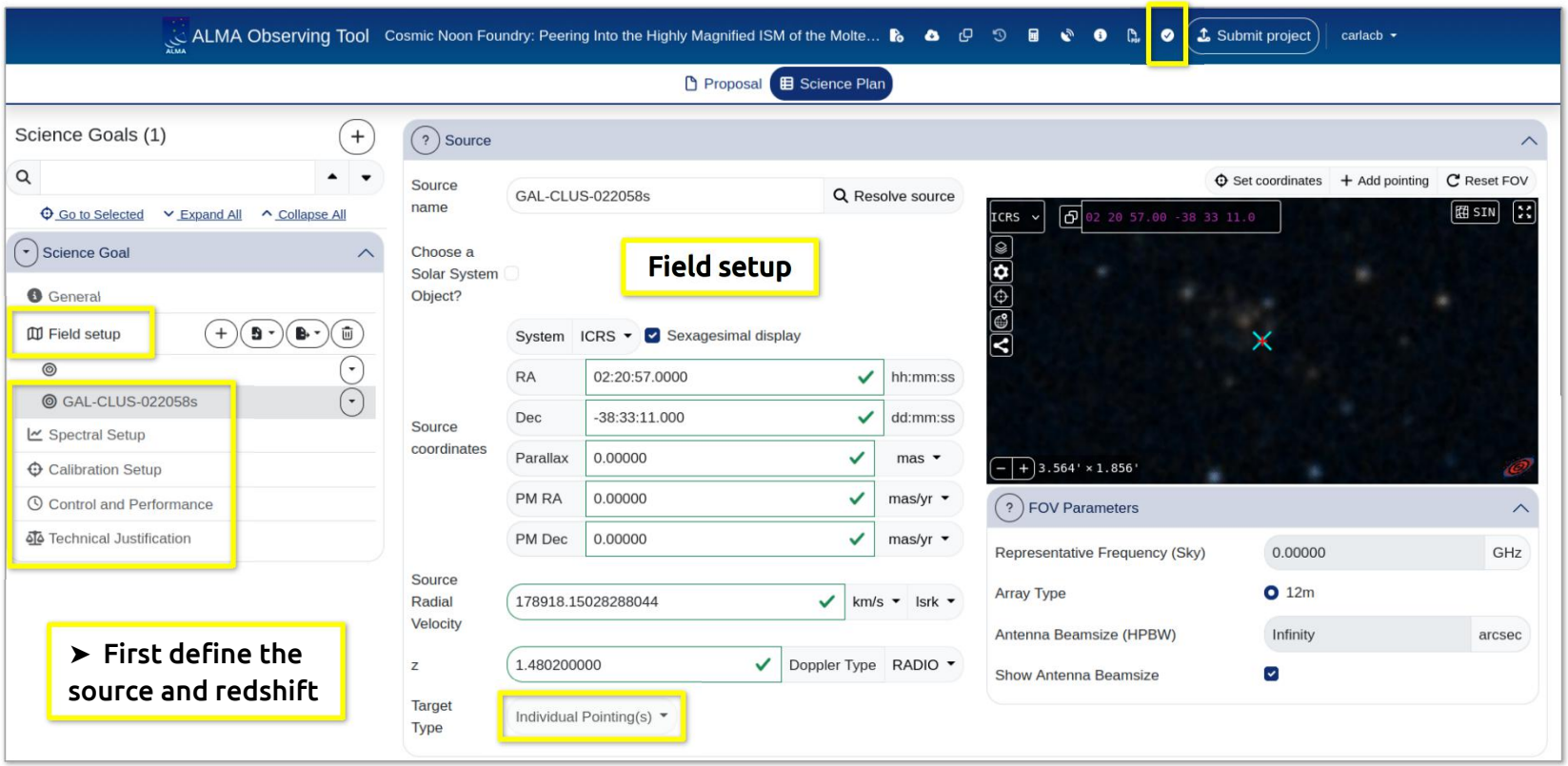
  - General info: title, abstract, type, science category
  - Joint Proposal (if any)
  - PI, Co-Is, Reviewer, Mentor
  - Science Justification (PDF)
  - Mention duplicate (if any)

- **Science Plan section**

  - Define your target(s)
  - Design the Science Goals

ALMA ngOT: https://cycle-13.sps.alma.cl/ngot/

New Web-Based OT



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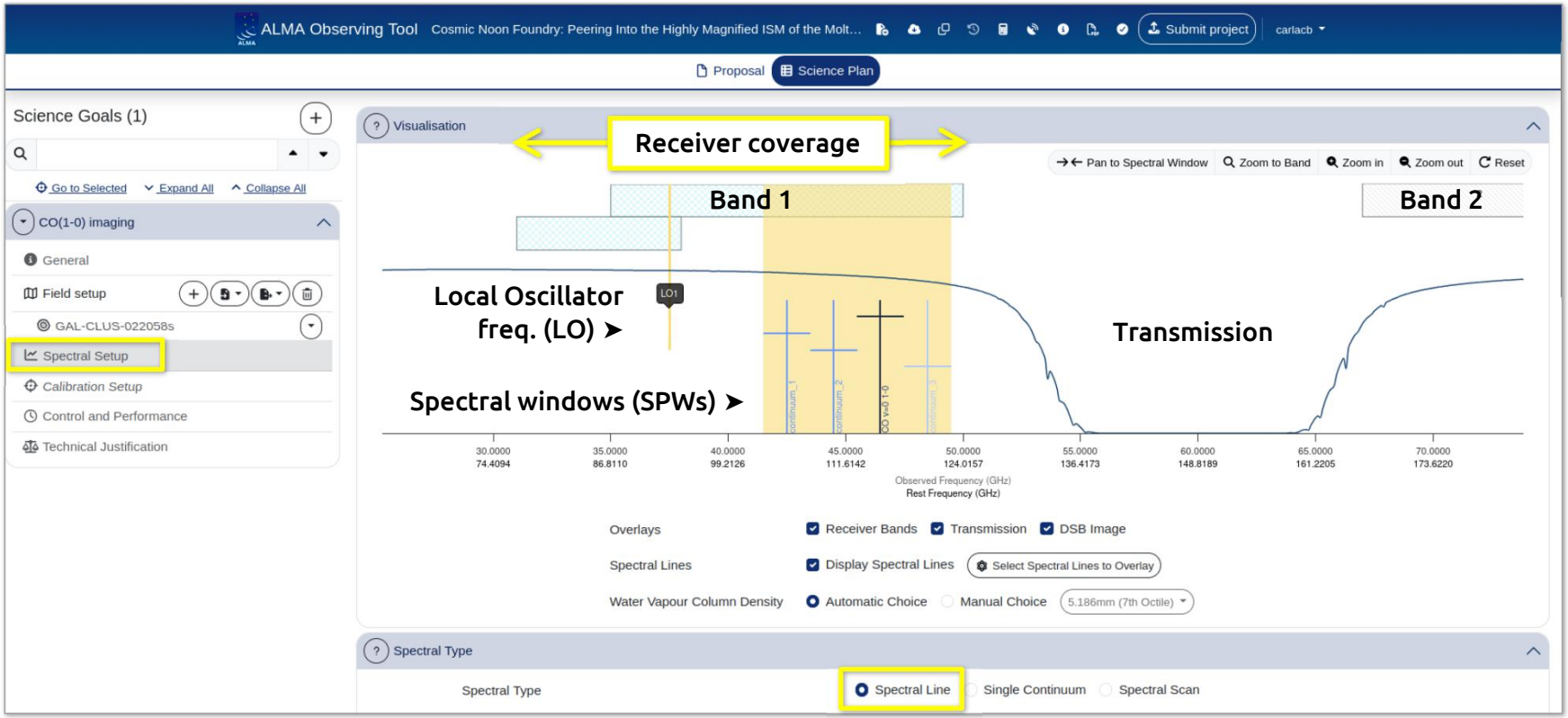


ALMA ngOT: https://cycle-13.sps.alma.cl/ngot/

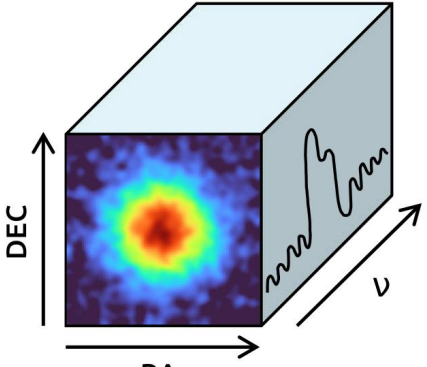
[Reminder] Frequency Sidebands

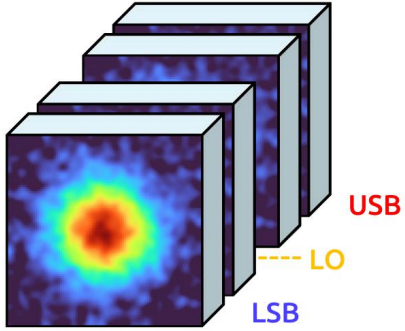
- Sky frequencies ~10^2 GHz
  - Too high to be processed directly by the electronics.
  - Sky signal is converted to a lower frequency (IF) using a very stable frequency (LO).

IF ≡ Intermediate Frequency
LO ≡ Local Oscillator

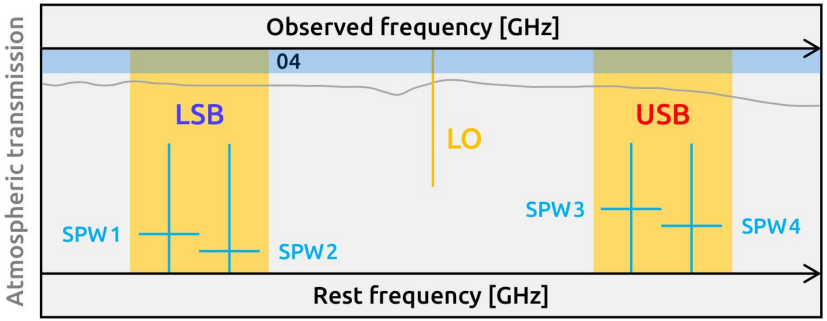
- Upper & Lower Sidebands
  - → USB = LO + IF
  - → LSB = LO – IF

Heterodyne principle: receiver bands produce two sidebands.

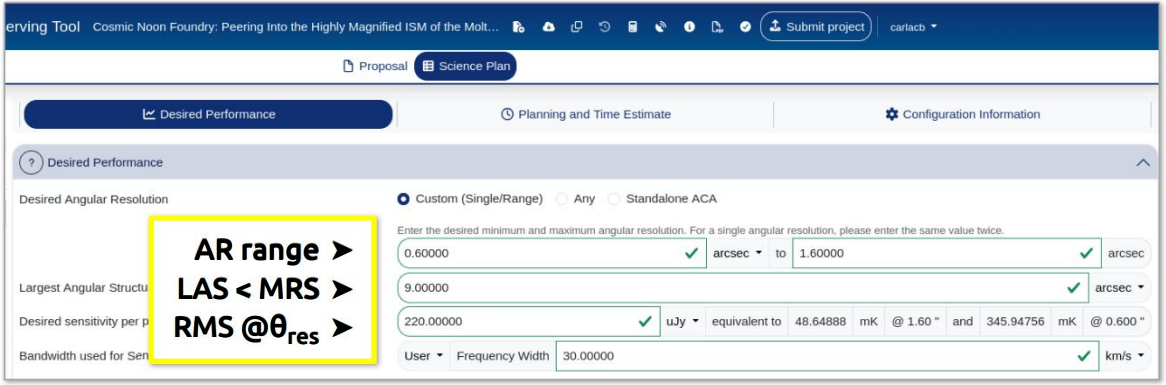


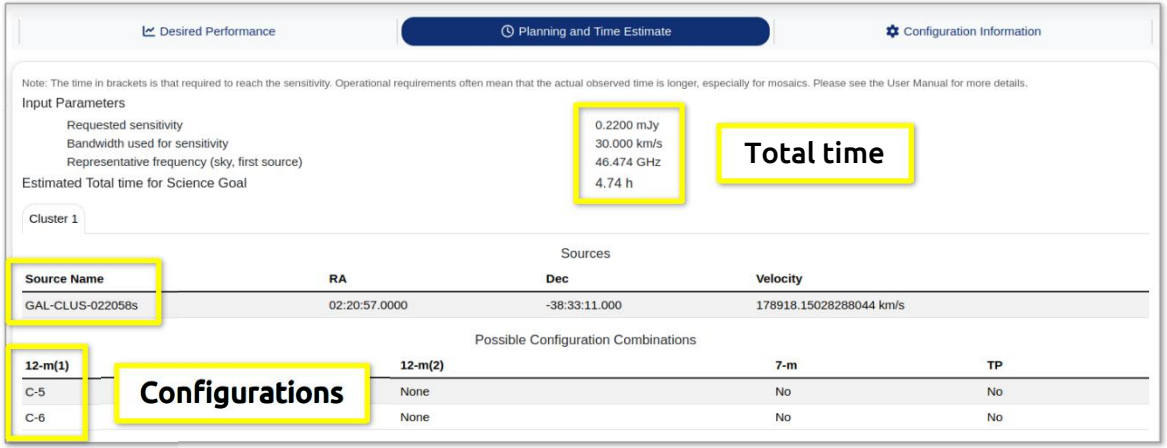


4 spectral windows



New Web-Based OT





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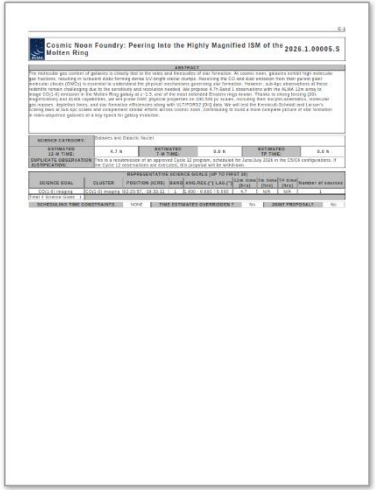
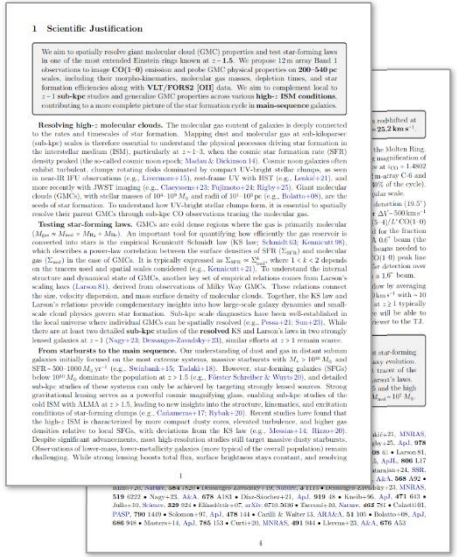
New Web-Based OT

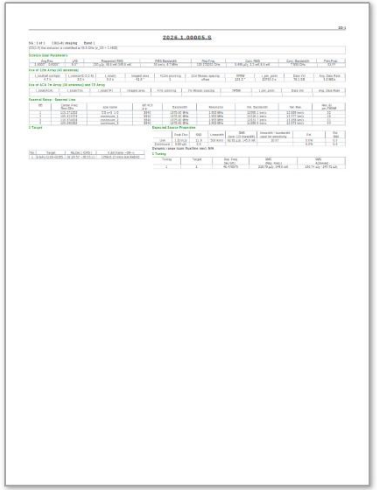
Validate → Generate a PDF of the whole proposal → Double-check → Submit



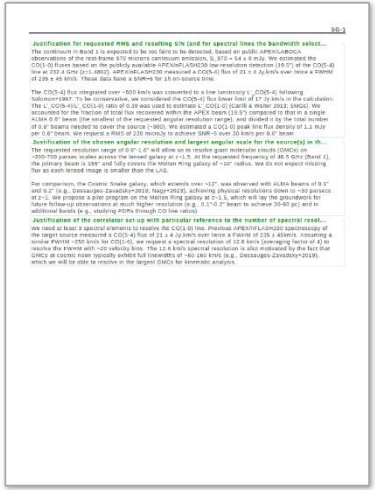
Proposal title, abstract, total time, SG list



Science Case (max 4 pages)



Science Goal summary



Technical Justification

➤ **After submission: Attribution of a Program ID + Email sent to the PI and Co-Is**

ALMA Observing Tool (OT)

Summary

- **Main differences from classic OT**
  - Web-based only (no local files)
  - Single version (autosave; each save overwrites)
  - Status: Draft / Submitted / Sub. with changes
  - Shared editing (Co-Is/Co-PIs)
- **Known issues/tricks**
  - No proposal templates
  - Switching pointing modes (Regular/Individual) or coords. systems can create many pointings
  - Band 2 continuum setup is limited
  - Loss of mosaic pointings when changing source
  - No “single” angular resolution option

Sources of Support

- **OT video tutorials**

  https://almascience.nrao.edu/proposing/observing-tool/video-tutorials
- **ALMA Helpdesk**

  Questions are usually answered within 48 hours

  Submit a ticket: https://help.almascience.org
- **Face-to-face Visitor Support**

  The NRAO/NAASC offers face-to-face meetings (online or in-person) for proposal preparation support and data reduction (PI data or archival)

  https://science.nrao.edu/facilities/alma/visitors-shortterm

ALMA ngOT: https://cycle-13.sps.alma.cl/ngot/